JavaScript 102 Project Plan | September 2026

Project direction: A simplified browser-based front end inspired by the existing LP Guard concentrated-liquidity project. The school project focuses on live market observation and range/composition analysis rather than wallet execution, blockchain writes, or autonomous rebalancing.

Website Name: LP Guard — Live Position Planner

Target User: Users who want to understand how a concentrated-liquidity price range affects the projected composition of an ETH/stablecoin position.

Website Purpose: Load live ETH market information, let the user define a lower and upper liquidity boundary, and clearly show range status plus estimated ETH/stablecoin exposure.

API: DEX Screener public API. The site uses the token-pairs endpoint for each network's canonical WETH token, then selects the most liquid direct WETH/USDC market on Base, Arbitrum, or Ethereum.

API Trigger: The user chooses a network, with USDC fixed as the quote asset for this project, then clicks "Load Live Data."

Live Data Displayed: USD price, 24-hour price change, 24-hour volume, USD liquidity, DEX/pair information, and a link to the market source when available.

Loading Message: "Loading live market data…" with a visible spinner and the selected market description.

No-Results Message: Explains that no matching WETH/USDC pair was found on the chosen network and asks the user to try another network or try again later.

Error Message: Explains that LP Guard could not reach the market-data service and asks the user to check the connection and try again.

Extra Sections: How It Works, About the Project / LP Guard modules, and a Feedback form.

AI Coding Agent: ChatGPT. It was used to translate a small part of the existing LP Guard work into beginner-appropriate JavaScript, plan the page, implement the API request and DOM interactions, simplify scope, test calculations, and review the final code.

Main Interactive Features

- **Live Market Snapshot:** Uses fetch(), async/await, response status checking, JSON parsing, try/catch, loading state, success state, no-results state, error state, and DOM updates.

- **Position Planner:** Accepts current price, lower range, upper range, and capital. It validates the range, determines Below Range / In Range / Above Range status, and estimates ETH versus stablecoin exposure.
- **Quick Range Controls:** Buttons set ±5%, ±10%, or ±15% boundaries around the current price.
- **Feedback Form:** Uses a submit event handler, prevents page refresh, checks required fields, displays confirmation, and clears the form after successful submission.

Scope Boundaries

- No wallet connection, private keys, signing, swaps, mints, removals, or other blockchain writes.
- No backend or database.
- No autonomous strategy selection or rebalancing.
- The result is educational decision support, not a transaction quote or financial advice.

Rough Layout Plan

Section	Contents
Header	LP Guard + navigation
Hero	Purpose + CTA buttons + read-only workflow summary
Live Market	Network control + fixed USDC quote + API result card
Position Planner	Price/range/capital form + analysis result
How It Works	Four-step explanation
About	Small explanation of Ledger / Risk / Optimize / Guard
Feedback	Name / email / message form
Footer	Course project label + DEX Screener credit

Reused LP Guard Concept

The planner reuses the earlier range/composition model: current price is compared with lower and upper boundaries, then the in-range composition is estimated with concentrated-liquidity square-root price math. Below the range the model becomes 100% ETH; above the range it becomes 100% stablecoin. This keeps the final project connected to work already completed while avoiding production LP Guard security and execution complexity.

Planned files: index.html, style.css, script.js, README.md, and this project-plan document.